Tri-Level Expansion Planning for Transmission Networks and Distributed Energy Resources Considering Transmission Cost Allocation

Jianxiao Wang, *Student Member, IEEE*, Haiwang Zhong, *Member, IEEE*, Wenyuan Tang, *Member, IEEE*, Ram Rajagopal, *Member, IEEE*, Qing Xia, *Senior Member, IEEE*, and Chongqing Kang, *Fellow, IEEE*

***Abstract*—A well-designed transmission cost allocation (TCA) scheme should reveal energy users' actual usage of transmission assets. By investing in distributed energy resources (DERs), a user should be allocated with a lower transmission price if devoted to reducing transmission capacity usage. To evaluate the impacts of TCA, a coplanning framework for centralized transmission networks and DERs is developed in this paper. Distinguished from the existing literature, we consider hourly transmission prices (TPs) that are updated by power flow tracing, together with locational marginal prices (LMPs) to incentivize users' investments for DERs. This problem is formulated as a tri-level model. On the first level, transmission expansion planning is optimized to satisfy transmission capacity requirements. On the second level, given the hourly LMPs and TPs, users at different buses strategically invest in DERs. On the third level, the market is cleared, and LMPs and TPs are updated. The lower two levels form a market equilibrium problem that can be solved by a diagonalization method. An iterative algorithm is proposed to solve the tri-level model, and convergence analysis is conducted. Case studies based on a modified Garver's 6-bus system and the IEEE 118-bus system demonstrate that a well-designed TCA scheme can incentivize users to invest in DERs, which effectively defers transmission expansion and reduces the system-wide investment costs.**

***Index Terms*—Distributed energy resource, market equilibrium, transmission cost allocation, transmission expansion planning.**

NOMENCLATURE

A. Indices and Sets

k	Line index.
s	Scenario index.
t	Time index.
x	Node index.
G	Superscript for conventional generators.
LD	Superscript for load demands.
LN	Superscript for transmission lines.
NLD	Superscript for net load demands.
Ω_x^G	Set of conventional generators at node x.
Ω^G	Set of conventional generators in the power system.
Ω^{LN}	Set of candidate and existing transmission lines.
Ω^{PLN}	Set of candidate transmission lines.
Ω^{ULN}	Set of existing lines and the lines to be constructed.

B. Parameters and Constants

B_k^{LN}	Admittance of transmission line k.
c_i^G	Unit cost of conventional generator i.
C_i^{ESS}	Capacity of ESS i.
$GSDF_{k-i}$	Generation shifting distribution factor of line k to node i.
$IC_i^{(\cdot)}$	Investment cost of DER i.
$IC_x^{\max}$	Investment budget at node x.
IC_k^{LN}	Investment cost of transmission line k.
N	Number of nodes in the power system.
$N^{(\cdot)}$	Number of DERs.
N^S	Number of scenarios.
$P_{(\cdot)}^{AWT}$	Available wind power.
$P_{(\cdot)}^{APV}$	Available solar power.
$P_{i,\alpha}^{ESS,\max}$	Maximal charging power of ESS i.
$P_{i,\beta}^{ESS,\max}$	Maximal discharging power of ESS i.
$P_{(\cdot)}^{LD}$	Load demand.
$P_k^{LN,\max}$	Transmission capacity of line k.
$P_i^{G,\min}$	Minimal power of conventional generator i.
$P_i^{G,\max}$	Maximal power of conventional generator i.
PTC_k	Transmission cost per unit power of line k.
$R(k)$	Receiving-end bus of transmission line k.
$S(k)$	Sending-end bus of transmission line k.
$SOC_i^{ESS,\min}$	Lower bound of the state of charge of ESS i.
$SOC_i^{ESS,\max}$	Upper bound of the state of charge of ESS i.
T	Number of time slots.
ΔT	Time interval.
γ_s	Weight of scenario s.

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J. Wang, H. Zhong, Q. Xia, and C. Kang are with the State Key Lab of Power Systems, Department of Electrical Engineering, Tsinghua University, Beijing 100084, China (e-mail: wang-jx14@mails.tsinghua.edu.cn; zhonghw@mail.tsinghua.edu.cn; qingxia@tsinghua.edu.cn; cqkang@tsinghua.edu.cn).

W. Tang is with the Department of Electrical and Computer Engineering, North Carolina State University, Raleigh, NC 27695 USA (e-mail: wtang8@ncsu.edu).

R. Rajagopal is with the Department of Civil and Environmental Engineering, Stanford University, Stanford, CA 94305 USA (e-mail: ramr@stanford.edu).

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$\eta_{i,\alpha}^{ESS}$ Charging efficiency of ESS *i*.
$\eta_{i,\beta}^{ESS}$ Discharging efficiency of ESS *i*.
$\delta_x^{\min}$ Minimal voltage phase angle of node *x*.
$\delta_x^{\max}$ Maximal voltage phase angle of node *x*.

C. Variables

C_x^{DER} Annualized investment in DERs at node *x*.
C_x^{LD} Annual cost of load demands at node *x*.
C^{TEP} Annual total costs for transmission expansion.
$E_{(\cdot)}^{ESS}$ Energy stored in the bank of ESS.
$P_{(\cdot)}^{NLD}$ Net load demands.
$P_{i,t,s,\alpha}^{ESS}$ Charging power of ESS *i* at time *t* in scenario *s*.
$P_{i,t,s,\beta}^{ESS}$ Discharging power of ESS *i* at time *t* in scenario *s*.
$P_{(\cdot)}^{G}$ Power of conventional generator.
$P_{(\cdot)}^{LN}$ Power on transmission line.
$P_{x,t,s}^{\text{total}}$ Total power extraction at node *x* at time *t* in scenario *s*.
$\mathbf{P}_{\mathbf{t,s}}^{\mathbf{total}}$ Vector of nodal total power extraction at time *t* in scenario *s*.
$\mathbf{P}_{\mathbf{t,s}}^{\mathbf{NLD}}$ Vector of net load demands at each node in the power system at time *t* in scenario *s*.
$\mathbf{P}_{\mathbf{t,s}}^{\mathbf{LN}}$ Vector of power flows on each line in the power system at time *t* in scenario *s*.
u_k^{LN} Binary variable of line *k*. A value of 1 indicates investing in line *k*. Otherwise 0.
$u_i^{(\cdot)}$ Binary variable of DER *i*. A value of 1 indicates investing in DER *i*. Otherwise 0.
$TC_{x,t,s}$ Transmission cost for node *x* at time *t* in scenario *s*.
$\lambda_{(\cdot)}^{E}$ Locational marginal price.
$\lambda_{(\cdot)}^{TC}$ Transmission price.
$\delta_{(\cdot)}$ Voltage phase angle.

I. Introduction

A. Motivation

TRANSMISSION costs, employed to recover the fixed costs of transmission assets, are critical components of users' electricity bills. In the Pennsylvania-New Jersey-Maryland (PJM) market, for example, the transmission price (TP) generally accounts for approximately 10% of the retail electricity price [1]. This value can reach 30% in the UK [2], and even higher in some other countries. A well-designed transmission cost allocation (TCA) mechanism can reveal energy users' actual usage of transmission assets, and incentivize users to make investments in distributed energy resources (DERs), thereby deferring transmission expansion. Therefore, it is imperative to develop a fair TCA scheme. However, in electricity markets across the world, transmission costs are generally determined annually based on the fixed costs and average usage of transmission assets [3]. In the United States, different regional transmission organizations (RTOs) have experimented with various transmission cost allocation schemes, summarized in Table I.

In Table I, the transmission costs for most energy users are determined according to the share of annual demand or summer peak demand, known as the postage stamp method [4].

TABLE I
TRANSMISSION COST ALLOCATION SCHEMES IN DIFFERENT RTOS

RTO	Responsibility	Method
CAISO	All loads	Share of annual demand [5]
ERCOT	All loads	Share of summer peak demand [6]
ISO-NE	All loads	Share of monthly coincident peak demand [7]
MISO	≥345 kV: 20% regionally and 80% sub-regionally 100–345 kV: sub-regionally	Regional loads: share of annual demand Sub-regional loads: line outage distribution factor analysis [8]
NYISO	All loads	Share of annual demand and contribution to reliability violation [9]
PJM	≥500 kV: all loads <500 kV and cost <5 million dollars: all loads <500 kV and cost ≥5 million dollars: direct beneficiaries	Share of annual demand [10]
SPP	≥300 kV: regionally 100–300 kV: 1/3 regionally and 2/3 zonally <100 kV: zonally	Share of annual demand [11]

Therefore, the transmission prices for energy users are predefined and constant, ignoring the impacts of time-varying load on the hourly actual usage of transmission assets.

A fair transmission cost allocation scheme should reveal users' hourly actual usage of transmission assets [2]. On the one hand, fair transmission prices should distinguish the contributions of user' hourly loads: A user should be allocated with a lower transmission price if devoted to reducing transmission capacity usage. On the other hand, fair transmission prices will justify DERs' costs and thereby incentivize the users' investment for DERs. In addition, with the increasing penetration of DERs, the peak loads of power systems, as well as the usage of transmission assets, can be effectively reduced. The rapid development of DERs is helpful to defer transmission expansion planning (TEP) while reducing the centralized investment costs. Therefore, it is imperative to incorporate a fair TCA scheme into the co-planning for transmission networks (TNs) and DERs, and comprehensively evaluate the grid-friendly benefits brought on by the fair TCA scheme.

Few existing studies investigate how TCA and time-varying transmission prices can influence distributed generation (DG) planning. In our previous work [2], we focus on the impacts of TCA on the planning strategy for DERs at a single bus. As an important component of electricity prices, a fair transmission price is able to boost the development of DERs while deferring the transmission expansion. There remain three major issues to be addressed: i) The impacts of TCA on the DERs planning across a power system; ii) The market equilibrium among different users considering TCA; iii) The grid-friendly benefits of TCA, e.g., how TCA helps defer transmission expansion. Therefore, we investigate the co-planning for TNs and DERs and quantify the grid-friendly benefits of TCA in this paper.

B. Literature Review and Contributions

The expansion planning for power networks and DERs have been widely investigated in past decades [12]. In [13], the

impacts of DG on transmission planning are evaluated by using a TEP model. The uncertainties in the system load and market prices are modeled via stochastic processes, which can be incorporated in the TEP model. In [14], a distribution system planning model considering DG is presented. In that model, optimal power flow is adopted to minimize capital costs for network upgrading, operation and maintenance costs, and the costs for load growth. In [15], a joint expansion planning problem of a distribution network and DG is addressed. The expansion plan optimizes the installed location and time for the candidate assets by minimizing the investment and operation costs. In [16], an active distribution network planning model is presented. The power supply reliability improvement brought on by ESSs is analyzed using that model. In [17], an approach of transmission network expansion considering N − 1 security constraints is proposed. In that work, the constraints for nodal short circuit currents are linearized and incorporated in the planning framework.

The multi-level planning framework has been widely applied to model the joint expansion for networks and DERs. In [18], a tri-level model of centralized transmission and DERs expansion planning is developed. In [19], a tri-level model is presented to co-optimize energy storage siting and sizing with transmission expansion. On the upper level, the lifetime profits of merchant energy storage are maximized. Then, TEP is optimized on the middle level and the market is cleared on the lower level. Among the existing literature, uncertainty modeling can be mainly divided into three types: i) scenarios, ii) uncertainty set, and iii) probabilistic constraints. In [20], a scenario-based multi-objective model is formulated for transmission expansion planning, in which load demands are modelled by typical scenarios. In [21], a robust transmission planning model is proposed considering wind power uncertainty. In that model, transmission expansion is robust against all scenarios that depict stochastic wind power. In [22], scenario-based stochastic programming is adopted to minimize the investment and expected operational costs of generation and network expansion planning. In [23], robust optimization is adopted for a resilient distribution network planning problem considering distributed generation resource placement. In [24], a chance-constrained optimization approach is proposed to tackle the uncertainty in load and wind power, which is used for transmission expansion planning.

Most of the existing literature focuses on minimizing the investment and operation costs of candidate assets based on predefined electricity prices [25], [26]. To the best of our knowledge, few studies have considered TCA in planning. To this end, this paper is *the first attempt* to co-plan centralized TNs and DERs considering the impacts of TCA. This problem is formulated as a tri-level model. On the first level, TEP is optimized to satisfy transmission capacity requirements. On the second level, given locational marginal prices (LMPs) and transmission prices, users at different buses respectively invest in DERs. On the third level, the hourly LMPs and TPs are updated based on direct current optimal power flow (DCOPF) and power flow tracing (PFT). By incorporating hourly TPs, the users will be incentivized to invest in DERs, which effectively defers transmission expansion. The contributions of this paper are as follows:

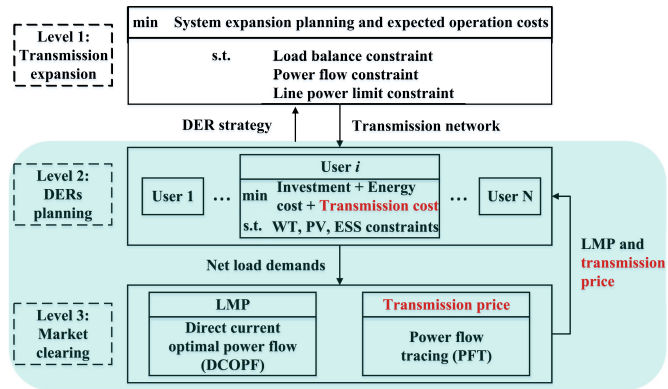

Fig. 1. The framework of the tri-level expansion planning considering TCA.

1) To our best knowledge, this is *the first work* to develop a tri-level planning model for TNs and DERs that considers TCA.
2) The impacts of TCA on deferring expansion planning for centralized TNs are investigated. A well-designed TCA scheme can incentivize users to invest in DERs, which substitutes transmission expansion and reduces the system-wide costs.
3) An iterative algorithm is proposed to solve the tri-level model, in which the market equilibrium problem formed by the lower two levels is approached by a diagonalization method.
4) The convergence analysis for the solution algorithm is conducted, and a criterion is designed to acquire a desired solution due to the nonlinearity of the tri-level model.

From the techno-economic analysis, TCA can incentivize users to invest in DERs, effectively deferring transmission expansion and reducing system-wide investment costs.

II. Framework

The framework of the tri-level expansion planning for TNs and DERs that considers TCA is shown in Fig. 1.

In Section III, the tri-level model is established. On the first level, transmission expansion planning model is formulated, which is to minimize the total investment and operation costs while satisfying load balance and transmission capacity requirements. On the second level, given LMPs and TPs, users located at different buses minimize individual costs by investing in DERs, including wind turbines (WTs), photovoltaic systems (PVs) and energy storage systems (ESSs). On the third level, hourly LMPs and TPs are updated by DCOPF and PFT. The users' planning strategies for DERs will change nodal net load, thereby influencing the LMPs and TPs. Thus, the lower two levels form a market equilibrium problem. In Section IV, an iterative algorithm is proposed to solve the tri-level model. Section V shows the case studies and Section VI concludes this paper.

The primary focus of this paper is to incorporate hourly TPs in the co-planning framework for TNs and DERs, and evaluate how a well-designed TCA scheme can incentive DERs investment while substituting centralized transmission expansion.

III. Tri-Level Model Formulation

In contrast to existing studies, hourly TPs that are updated by PFT, together with LMPs are generated to incentivize users' investments for DERs in the proposed tri-level model.

A. First Level

On the first level, transmission expansion planning model is established to minimize the annualized expansion planning and expected annual operation costs while satisfying transmission capacity requirements. The TEP model is as follows:

$$\min\ C^{TEP} = \sum_{k\in\Omega^{PLN}} u_k^{LN} IC_k^{LN} + \sum_{s=1}^{N^S}\gamma_s \sum_{t=1}^{T}\sum_{i\in\Omega^G} c_i^G P_{i,t,s}^G, \tag{1}$$

subject to

$$\sum_{i\in\Omega_x^G} P_{i,t,s}^G - P_{x,t,s}^{NLD} = \sum_{k|S(k)=x} P_{k,t,s}^{LN} - \sum_{k|R(k)=x} P_{k,t,s}^{LN},\ \forall x,t,s, \tag{2}$$

$$P_{k,t,s}^{LN} - B_k^{LN}\left(\delta_{S(k),t,s} - \delta_{R(k),t,s}\right) = 0,\ \forall t,s,k\in\Omega^{LN}\backslash\Omega^{PLN}, \tag{3}$$

$$-P_k^{LN,\max} \le P_{k,t,s}^{LN} \le P_k^{LN,\max},\ \forall t,s,k\in\Omega^{LN}\backslash\Omega^{PLN}, \tag{4}$$

$$P_{k,t,s}^{LN} - B_k^{LN}\left(\delta_{S(k),t,s} - \delta_{R(k),t,s}\right) + \left(1-u_k^{LN}\right)\mathrm{M} \ge 0, \forall t,s,k\in\Omega^{PLN}, \tag{5}$$

$$P_{k,t,s}^{LN} - B_k^{LN}\left(\delta_{S(k),t,s} - \delta_{R(k),t,s}\right) - \left(1-u_k^{LN}\right)\mathrm{M} \le 0, \forall t,s,k\in\Omega^{PLN}, \tag{6}$$

$$-u_k^{LN}P_k^{LN,\max} \le P_{k,t,s}^{LN} \le u_k^{LN}P_k^{LN,\max},\ \forall t,s,k\in\Omega^{PLN}, \tag{7}$$

$$\delta_x^{\min} \le \delta_{x,t,s} \le \delta_x^{\max},\ \forall x,t,s, \tag{8}$$

$$P_i^{G,\min} \le P_{i,t,s}^G \le P_i^{G,\max},\ \forall i,t,s. \tag{9}$$

$$u_k^{LN} \in \{0,1\},\ \forall k. \tag{10}$$

In this model, (2) is the power balance constraint. (3) and (4) are the power flow constraints for existing lines. (5)–(7) are the power flow constraints for candidate lines, where M is a large number greater than $B_k^{LN}(\delta_{S(k)}^{\max} - \delta_{R(k)}^{\min})$. (8) and (9) show the limits for voltage phase angles and generator outputs, respectively. Note that the impacts of long-term load increase and reserve requirement can be considered in this model by generating the associated load scenarios in (2).

B. Second Level

On the second level, different users located across the power system minimize individual costs by investing in DERs. Here, the users refer to large consumers in the transmission network, e.g., industrial parks or commercial buildings, so the users' investments for DERs can influence LMPs and TPs. Given LMPs and TPs, the model of the user at node *x* is as follows:

$$\min\ C_x^{DER} + C_x^{LD}, \tag{11}$$

subject to

$$C_x^{DER} = \sum_{i=1}^{N^{WT}} u_i^{WT} IC_i^{WT} + \sum_{i=1}^{N^{PV}} u_i^{PV} IC_i^{PV} + \sum_{i=1}^{N^{ESS}} u_i^{ESS} IC_i^{ESS} \le IC_x^{\max}, \tag{12}$$

$$C_x^{LD} = \sum_{s=1}^{N^S}\gamma_s \sum_{t=1}^{T}\left(\lambda_{x,t,s}^E + \lambda_{x,t,s}^{TC}\right) P_{x,t,s}^{NLD}\Delta T, \tag{13}$$

$$0 \le P_{i,t,s}^{WT} \le u_i^{WT} P_{i,t,s}^{AWT},\ \forall i,t,s, \tag{14}$$

$$0 \le P_{i,t,s}^{PV} \le u_i^{PV} P_{i,t,s}^{APV},\ \forall i,t,s, \tag{15}$$

$$0 \le P_{i,t,s,\alpha}^{ESS} \le u_i^{ESS} P_{i,\alpha}^{ESS,\max},\ \forall i,t,s, \tag{16}$$

$$0 \le P_{i,t,s,\beta}^{ESS} \le u_i^{ESS} P_{i,\beta}^{ESS,\max},\ \forall i,t,s, \tag{17}$$

$$E_{i,t,s}^{ESS} = E_{i,t-1,s}^{ESS} + \left(P_{i,t,s,\alpha}^{ESS}\eta_{i,\alpha}^{ESS} - P_{i,t,s,\beta}^{ESS}/\eta_{i,\beta}^{ESS}\right) \times \Delta T,\ \forall i,t,s, \tag{18}$$

$$SOC_i^{ESS,\min} \le E_{i,t,s}^{ESS}/C_i^{ESS} \le SOC_i^{ESS,\max},\ \forall i,t,s, \tag{19}$$

$$E_{i,0,s}^{ESS} = E_{i,T,s}^{ESS},\ \forall i,s, \tag{20}$$

$$P_{x,t,s}^{NLD} = P_{x,t,s}^{LD} - \sum_{i=1}^{N^{WT}} P_{i,t,s}^{WT} - \sum_{i=1}^{N^{PV}} P_{i,t,s}^{PV} - \sum_{i=1}^{N^{ESS}}\left(-P_{i,t,s,\alpha}^{ESS} + P_{i,t,s,\beta}^{ESS}\right),\ \forall t,s. \tag{21}$$

$$u_i^{WT}, u_i^{PV}, u_i^{ESS} \in \{0,1\},\ \forall i. \tag{22}$$

In this model, (12) shows that the investment costs including WTs, PVs and ESSs, are restricted by the budget. In (13), the electricity costs are composed of energy and transmission costs. The LMPs and TPs in (13) are influenced by the users' planning strategy. Note that the investment costs of DERs are annualized, and the electricity costs are in annual values. (14) and (15) show that the generation of WTs and PVs is limited by the available power at each time slot. (16) and (17) show the limit for the charging and discharging power of ESSs. (18) shows the dynamic process of the energy stored in the bank of ESSs. (19) shows that the stored energy is bounded by the state of charge (SOC). In (20), the stored energy at time *T* is set to be the same as the initial value. (21) shows that the net load is equal to the difference between the load and DERs' output.

In this model, we assume that the variable costs for DERs are not considered as the existing literature does [18], [19]. To evaluate the impacts of DERs' variable costs on the planning strategies, a sensitivity analysis is conducted in the case studies.

In addition, to focus on the interactions between the power system and large consumers, distribution networks are ignored.

C. Third Level

The market is cleared on the third level. LMPs are updated by DCOPF, and TPs are updated using PFT.

1) LMP Model: DCOPF has been widely used for computing LMPs in existing research and electricity markets, e.g., the Electric Reliability Council of Texas. In DCOPF, the power losses are not considered for simplification. $\forall s, t$, the DCOPF model is:

$$\min \sum_{i\in\Omega^G} c_i^G \times P_{i,t,s}^G, \tag{23}$$

subject to

$$\sum_{i\in\Omega^G} P_{i,t,s}^G = \sum_{i\in\Omega^{LD}} P_{i,t,s}^{NLD} : \lambda_{t,s}, \tag{24}$$

$$\sum_{i=1}^{N} GSDF_{k-i}\left(P_{i,t,s}^G - P_{i,t,s}^{NLD}\right) \le P_k^{LN,\max} : \mu_{k,t,s}^+,\ \forall k \in \Omega^{ULN}, \tag{25}$$

$$\sum_{i=1}^{N} GSDF_{k-i}\left(P_{i,t,s}^G - P_{i,t,s}^{NLD}\right) \ge -P_k^{LN,\max} : \mu_{k,t,s}^-,\ \forall k \in \Omega^{ULN}, \tag{26}$$

$$P_i^{G,\min} \le P_{i,t,s}^G \le P_i^{G,\max},\ \forall i. \tag{27}$$

Note that Ω^{ULN} is updated with the TEP results. When the planning strategies for TNs and DERs are determined, the DCOPF model is to minimize the operation costs of conventional generators in the power system, subject to power balance constraint (24), line power flow constraints (25) and (26), and the limits for generator output (27). The Lagrangian multipliers of constraints (24)–(26) are denoted by $\lambda_{t,s}$, $\mu_{k,t,s}^+$ and $\mu_{k,t,s}^-$, respectively. Then the LMP at node x, $\lambda_{x,t,s}^E$, is:

$$\lambda_{x,t,s}^E = \lambda_{t,s} + \sum_{k\in\Omega^{ULN}} GSDF_{k-x}\left(\mu_{k,t,s}^+ + \mu_{k,t,s}^-\right). \tag{28}$$

2) TCA Model: In this paper, all transmission costs are assumed to be recovered by energy users [4]. The PFT method is adopted to allocate the transmission costs to each user [27].

The power flows are determined by using DCOPF. Then, the total power extraction at node x is calculated as the sum of the net load and the transmission power from node x to other nodes:

$$P_{x,t,s}^{\text{total}} = \sum_{k|S(k)=x} P_{k,t,s}^{LN} + P_{x,t,s}^{NLD}. \tag{29}$$

To obtain the relationship between the total power extraction and the net load, (29) is derived as follows:

$$P_{x,t,s}^{\text{total}} - \sum_{k|S(k)=x} \left(P_{k,t,s}^{LN}/P_{y,t,s}^{\text{total}}\right) \cdot P_{y,t,s}^{\text{total}}|_{y=R(k)} = P_{x,t,s}^{NLD}. \tag{30}$$

The matrix form is expressed as follows:

$$\mathbf{A_{t,s}P_{t,s}^{total}} = \mathbf{P_{t,s}^{NLD}}, \tag{31}$$

where $\mathbf{A_{t,s}}$ is the relationship matrix, which can be derived from the power flows after DCOPF. The element a_{xy} of $\mathbf{A_{t,s}}$ in row x and column y is:

$$a_{xy} = \begin{cases} 1, & x = y \\ -P_{k,t,s}^{LN}/P_{y,t,s}^{\text{total}}, & S(k) = x,\ R(k) = y \\ 0, & \text{otherwise.} \end{cases} \tag{32}$$

Since $\mathbf{A_{t,s}}$ is invertible,

$$\mathbf{P_{t,s}^{total}} = \mathbf{A_{t,s}^{-1}P_{t,s}^{NLD}}. \tag{33}$$

The matrix $\mathbf{A_{t,s}^{-1}}$ represents the contributions of the net loads to the nodal total output power. The relationship between the line power flow and the nodal total power extraction is:

$$\mathbf{P_{t,s}^{LN}} = \mathbf{C_{t,s}P_{t,s}^{total}}. \tag{34}$$

According to the *proportional sharing principle* [4], the element c_{kx} of the matrix $\mathbf{C_{t,s}}$ in row k and column x is:

$$c_{kx} = \begin{cases} P_{k,t,s}^{LN}/P_{x,t,s}^{\text{total}}, & R(k) = x \\ 0, & \text{otherwise.} \end{cases} \tag{35}$$

According to (33) and (34), the relationship between the line power flow and the net load is:

$$\mathbf{P_{t,s}^{LN}} = \mathbf{C_{t,s}A_{t,s}^{-1}P_{t,s}^{NLD}} = \mathbf{F_{t,s}P_{t,s}^{NLD}}, \tag{36}$$

where the matrix $\mathbf{F_{t,s}}$ represents the contributions of the net loads to the line power flows. Then, the transmission costs of all transmission lines allocated to node x are:

$$TC_{x,t,s} = \sum_{k\in\Omega^{ULN}} PTC_k \cdot f_{kx} \cdot P_{x,t,s}^{NLD}, \tag{37}$$

where f_{kx} is the element of the matrix $\mathbf{F_{t,s}}$ in row k and column x. $f_{kx}P_{x,t,s}^{NLD}$ represents the power flow on line k contributed by the net load at node x. Hence, the transmission price $\lambda_{x,t,s}^{TC}$ is:

$$\lambda_{x,t,s}^{TC} = \frac{TC_{x,t,s}}{P_{x,t,s}^{NLD}} = \sum_{k\in\Omega^{ULN}} PTC_k \cdot f_{kx}. \tag{38}$$

IV. Solution Algorithm

As aforementioned, the lower two levels, i.e., the interactions between users and market operator, form a market equilibrium problem. As there are multiple users, this problem is an equilibrium problem with equilibrium constraints (EPEC). We firstly solve the EPEC by a diagonalization method in Section IV-A, and then solve the interactions between transmission expansion and EPEC by an iterative algorithm in Section IV-B. In Section IV-C, the convergence analysis is conducted and a criterion is presented to obtain a desired solution to the tri-level model.

A. Diagonalization Method

There has been a number of literature focused on the solution algorithm of EPEC. In [18], two approaches are described

to find an equilibrium solution if one exists. One approach is the diagonalization method, which iteratively solves the EPEC model until a stationary point is obtained. The other is the nonlinear complementarity problem (NCP), which combines the Karush-Kuhn-Tucker (KKT) conditions of each equilibrium sub-problem to reformulate the EPEC model into a mixed complementarity problem (MCP). Then the MCP is solved as a single-level optimization model. In [28], another approach termed as the potential function is applied. By introducing the potential function, the EPEC model can be equivalently transformed into a linearized optimization model. Note that the potential function approach only works for a potential game [29].

In this paper, a diagonalization method [30] is adopted to solve the EPEC formed by the lower two levels. The diagonalization method is to iteratively solve each user's DER planning model considering market clearing with other users' strategies fixed. Each user's planning model and the market clearing model form a market problem with equilibrium constraints (MPEC). After solving all MPECs, one iteration is completed. The iterations continue until users' planning strategies converge.

Diagonalization Method.

Initialize the iteration index $k = 0$, all users' planning strategies, i.e., $u_i^m[0]=0,\ \forall i,\ m\in\{WT,\ PV,\ ESS\}$, and the net load at node *x*, i.e., $P_{x,t,s}^{NLD}[0] = P_{x,t,s}^{LD}$.
Repeat
  For each user *x*, fix other users' planning strategies and nodal net load.
  Repeat
    User *x* optimizes his/her DER planning model (11)–(22).
    The market operator updates LMPs by optimizing (23)–(28), and TPs by solving (36)–(38).
  Until user *x*'s planning strategy converges.
  Update user *x*'s planning strategy, i.e., $u_i^m[k\ +\ 1],\ \forall i,\ m \in \{WT,\ PV,\ ESS\}$, and nodal net load, i.e., $P_{x,t,s}^{NLD}[k\ +\ 1]$.
  After optimizing all users' interactions with the market operator, $k = k + 1$.
Until all users' planning strategies converge, i.e., for each user *x*, $u_i^m[k\ +\ 1] = u_i^m[k],\ \forall i,\ m \in \{WT,\ PV,\ ESS\}$.

B. Iterative Algorithm

The centralized transmission expansion on the first level and the users' planning for DERs on the second level are influenced by each other. On the one hand, constructing more transmission lines will increase the TPs for users, which incentivizes the users to use DERs instead of electricity from power grids. On the other hand, the users' investment for DERs will reduce nodal net load in power grids, which lowers transmission capacity requirement and defers transmission expansion. To balance this trade-off, an iterative algorithm is proposed in Fig. 2.

The iterative algorithm is to coordinate the centralized transmission expansion planning and the EPEC formed by the lower two levels. Firstly, solve EPEC to get all users' planning strategies for DERs, and update the nodal net load. Secondly, based on the updated net load, the TEP model is optimized, and transmission networks are updated. These two steps iterate until the decisions for transmission networks converge or the iteration index exceeds the pre-defined limit.

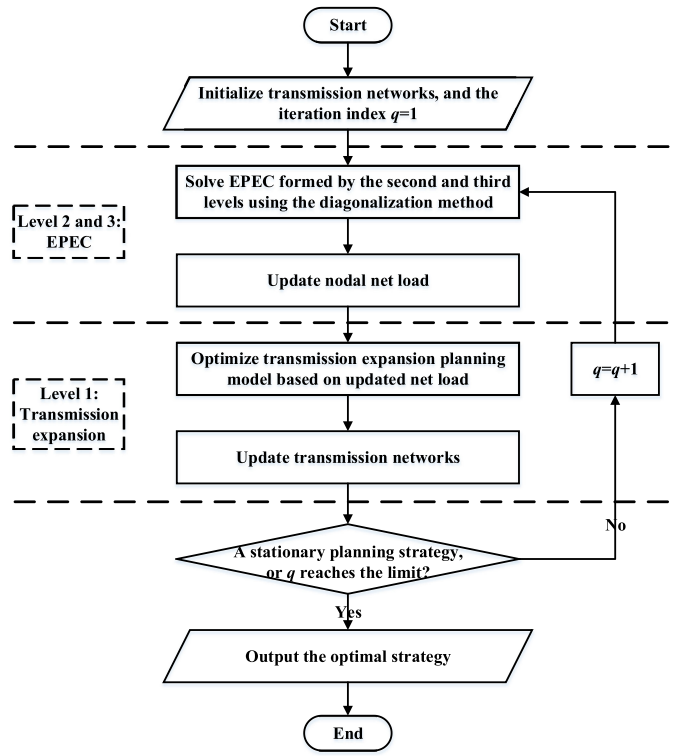

Fig. 2. The iterative algorithm to solve the tri-level model.

C. Convergence Analysis

The tri-level model can be reformulated into a single-level optimization program by integrating the first-order optimality conditions of EPEC into the TEP model. This single-level model is a mixed integer nonlinear programming (MINLP) problem. Due to the nonlinearity and concavity of the tri-level model, the optimality and convergence of the proposed iterative algorithm cannot be always guaranteed. As explained in [18], the planning strategies for TNs and DERs may oscillate, usually among two or more different decisions because of possible occurrence of non-existing pure Nash equilibrium solution.

The aforementioned oscillations are illustrated in a dilemma between two situations: i) The users invest in the marginal DERs, replacing the marginal transmission line. As the electricity and transmission prices vary with the planning strategies for TNs and DERs, the users will possibly find it not beneficial to invest in the marginal DERs. ii) The users choose not to invest in the marginal DERs, leading to the expansion for the marginal transmission line. Then the users find it beneficial to use the marginal DERs instead of electricity from power systems. The dilemma is illustrated in Fig. 3, where the marginal DER is denoted by DER_n and the marginal transmission line is TN_n.

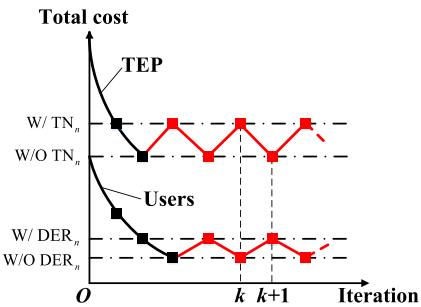

Fig. 3. Illustration of the iterative oscillation.

From the perspective of the users and transmission operator, we give a theoretical analysis for judging whether the oscillations will occur or not. For user x, let $\lambda[k]$ represent the LMPs and TPs in the kth iteration, $\mathbf{U}_{\mathbf{x}}^{\mathbf{DER}}[k+1]$ represent the optimal planning strategy for DERs under $\lambda[k]$, and $\mathbf{P}_{\mathbf{x}}^{\mathbf{NLD}}[k]$ represent the optimal nodal net load under $\mathbf{U}_{\mathbf{x}}^{\mathbf{DER}}[k]$.

Theorem 1: $\forall k \in \mathbf{N}^*$, user x's planning strategy for DERs oscillate if and only if

$$\begin{aligned} &C_x^{DER}\left(\mathbf{U}_{\mathbf{x}}^{\mathbf{DER}}[k]\right) + C_x^{LD}\left(\lambda[k], \mathbf{P}_{\mathbf{x}}^{\mathbf{NLD}}[k]\right) \\ &\quad > C_x^{DER}\left(\mathbf{U}_{\mathbf{x}}^{\mathbf{DER}}[k+1]\right) \\ &\qquad + C_x^{LD}\left(\lambda[k], \mathbf{P}_{\mathbf{x}}^{\mathbf{NLD}}[k+1]\right). \end{aligned} \tag{39}$$

Theorem 1 can be interpreted as follows: Given the kth LMPs and TPs $\lambda[k]$, the total cost for the strategy $\mathbf{U}_{\mathbf{x}}^{\mathbf{DER}}[k]$ is greater than that for the strategy $\mathbf{U}_{\mathbf{x}}^{\mathbf{DER}}[k+1]$, i.e., the optimal planning strategy under $\lambda[k]$.

For the transmission operator, let $\mathbf{P}^{\mathbf{NLD}}[k]$ represent the nodal net load in the power system in the kth iteration and $\mathbf{U}^{\mathbf{TEP}}[k+1]$ represent the optimal planning strategy for TNs under $\mathbf{P}^{\mathbf{NLD}}[k]$.

Theorem 2: $\forall k \in \mathbf{N}^*$, the planning strategy for TNs oscillate if and only if

$$\begin{aligned} &C^{TEP}\left(\mathbf{U}^{\mathbf{TEP}}[k], \mathbf{P}^{\mathbf{NLD}}[k]\right) \\ &\quad > C^{TEP}\left(\mathbf{U}^{\mathbf{TEP}}[k+1], \mathbf{P}^{\mathbf{NLD}}[k]\right). \end{aligned} \tag{40}$$

Similarly, the planning strategy for TNs will not converge if given the kth net load $\mathbf{P}^{\mathbf{NLD}}[k]$, the total cost for the strategy $\mathbf{U}^{\mathbf{TEP}}[k]$ is greater than that for the strategy $\mathbf{U}^{\mathbf{TEP}}[k+1]$.

In this paper, a criterion is designed to get the desired solution to the tri-level model if the planning strategies of the users and transmission operator oscillate.

The main idea of the proposed criterion is to search for the situation with minimal system-wide total costs among the oscillations. The LMPs and TPs are used for settlement. The convergence analysis will be demonstrated in our case studies.

Criterion for desired solution.

1. Find all the situations where oscillations occur during the iterations;
2. In each oscillated situation, calculate the LMPs and TPs based on the planning strategies for TNs and DERs;
3. Apply the LMPs and TPs for settlement: Calculate the users' energy and transmission costs;
4. Calculate the system-wide total costs for each oscillated situation, including the users' investments for DERs, the users' energy and transmission costs, the investment for TNs, and the generation costs.
5. Find the situation with minimal system-wide total costs, and the planning strategies for DERs and TNs in the selected situation are the desired solutions.

V. Case Studies

The program is developed using MATLAB R2015b operating at 2.40 GHz with 8 cores. The optimization solver is CPLEX 12.4 [31]. In this section, a modified Garver's 6-bus system and the IEEE 118-bus system are simulated to validate the proposed model. According to the reports from World Energy Council [32]–[34], the capital costs for WTs and PVs are around 1.5 and 0.8 dollars/W. To derive annualized investment costs, we apply the capital recovery factor $r(1+r)^{PP}/[(1+r)^{PP}-1]$ to the investment costs, where r is the interest rate, i.e., 1.25% [35], and PP is the payback period, i.e., 20 years. There has been a number of literature demonstrating that it is not profitable to invest in storage under current pricing schemes due to the high capital costs [36]. Therefore, we set the capital costs for ESSs to 90 dollars/kW. Our results offer an analysis for storage investment in the future when the investment costs are low. In [37], storage prices are projected to reduce by 30% by 2020. The parameters of DERs are listed in Table II. Note that the investment costs are annualized.

TABLE II
Parameters of DERs

DER	Investment cost ($)	Capacity (MW)	
WT	5.0×10^5	6	
PV	1.8×10^5	4	
ESS	Investment cost ($)	$SOC_{i,\min}^{ESS}$	$SOC_{i,\max}^{ESS}$
	1.0×10^4	10%	90%
	C_i^{ESS} (MWh)	$\eta_{i,\alpha}^{ESS}$	$\eta_{i,\beta}^{ESS}$
	10	95%	95%
	$E_{i,0}^{ESS}$ (MWh)	$P_{i,\alpha,\max}^{ESS}$ (MW)	$P_{i,\beta,\max}^{ESS}$ (MW)
	3	2	2

To address the uncertainties in load and renewable energy, scenario-based stochastic programming is adopted. The yearly wind, photovoltaic and load power curves from a realistic dataset in China are clustered into 10 typical daily scenarios via K-means clustering. The transmission cost for each line is 10 $/h for a 1 MW capacity, i.e., $PTC_k = 10,\ \forall k$.

A. Modified Garver's 6-Bus System

The modified Garver's 6-bus system is shown in Fig. 4.

The parameters of the centralized thermal units and the transmission lines are listed in Tables III and IV, respectively [35].

Two users, located at nodes 4 and 5, make investments in DERs. Each user has an annualized budget of $2,000,000 for candidate DERs, including 3 WTs, 3 PVs and 2 ESSs. To demon-

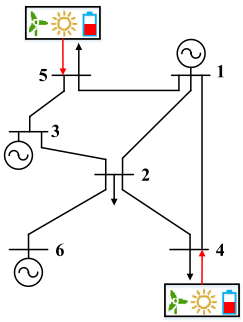

Fig. 4. The modified Garver's 6-bus system.

TABLE III
PARAMETERS OF CENTRALIZED THERMAL UNITS

Unit No.	Node	$P^{G,\max}$ (MW)	Cost ($/MWh)
1	1	30	10
2	3	20	40
3	6	50	70

TABLE IV
PARAMETERS OF TRANSMISSION LINES

Line No.	From	To	Reactance (p.u.)	$P^{LN,\max}$ (MW)	Status	Investment (million $)
1	1	2	0.4	10	Existing	/
2	1	4	0.6	8	Existing	/
3	1	5	0.2	15	Existing	/
4	2	3	0.2	15	Existing	/
5	2	4	0.4	10	Existing	/
6	2	6	0.3	10	Existing	/
7	3	5	0.2	10	Existing	/
8	2	4	0.4	10	Candidate	12
9	2	6	0.3	10	Candidate	10
10	2	6	0.3	10	Candidate	10
11	3	5	0.2	10	Candidate	10
12	4	6	0.3	6	Candidate	11
13	4	6	0.3	6	Candidate	11
14	4	6	0.3	6	Candidate	11

strate the impacts of TCA on the planning strategies of TNs and DERs, three scenarios are adopted: i) S1, where transmission expansion planning is optimized without considering DERs; ii) S2, where DERs are planned for while only considering LMPs, and then transmission expansion planning is optimized; and iii) S3, the proposed method, where DERs are planned for while considering both LMPs and transmission prices, and then transmission expansion planning is optimized.

By comparing the performance of S1 and S3, the impacts of DERs on transmission expansion planning can be demonstrated. By comparing the performance of S2 and S3, the impacts of TCA on boosting the DER investments and deferring transmission expansion can be illustrated.

1) Optimal Planning Strategy in S3: The planning strategies for DERs and TNs are shown in Table V. A value of 1 indicates that the line is optimized for construction, while 0 indicates the line will not be constructed.

TABLE V
OPTIMAL PLANNING STRATEGIES FOR DERS AND TNS IN S3

DER	WT		PV		ESS		
Node 4	3		2		2		
Node 5	2		0		2		
Line No.	8	9	10	11	12	13	14
Result	1	1	1	0	0	0	0

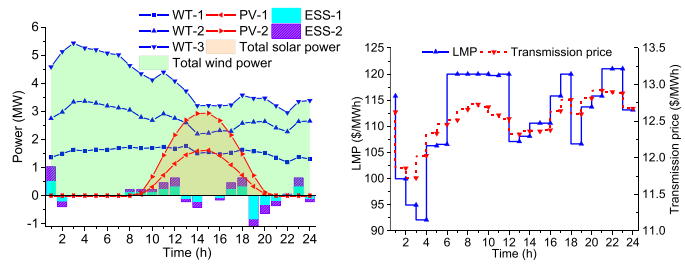

Fig. 5. Expected power curves and prices at node 4.

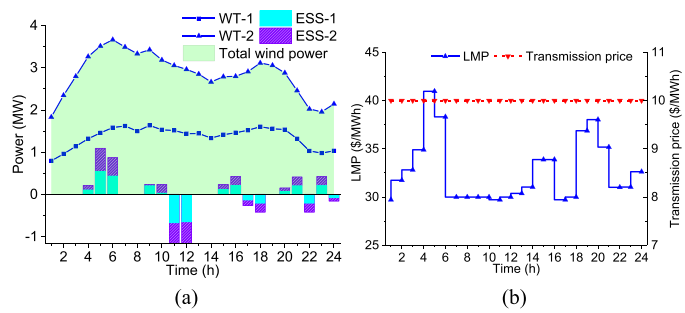

Fig. 6. Expected power curves and prices at node 5.

In S3, the users at node 4 will invest in 3 WTs, 2 PVs and 2 ESSs, while the users at node 5 will invest in 2 WTs and 2 ESSs. In addition, 3 transmission lines will be constructed to satisfy the transmission capacity requirements. Figs. 5(a) and 6(a) show the expected power curves of the invested DERs at nodes 4 and 5. Figs. 5(b) and 6(b) show the expected LMPs and transmission prices at nodes 4 and 5.

From the results, the wind and solar power are fully accommodated because the operation costs of WTs and PVs are 0. Driven by LMPs and transmission prices, the ESSs charge during valley hours and discharge during peak hours. For example, the user's ESSs at node 4 charge from 19:00 to 21:00 when prices are low, and discharge from 8:00 to 12:00 when prices are high. By means of arbitrage, the ESSs can effectively reduce users' costs by shifting loads.

2) Comparisons Between Different Scenarios: The planning strategies for DERs in three scenarios are shown in Table VI. The planning strategies for TNs in three scenarios are compared in Table VII.

From the results in Table VI, transmission prices can further encourage users to invest in DERs. In this case, the user at node 5 invests in 2 more WTs in S3 compared with the results in S2. From the results in Table VII, DERs can substitute transmission lines as an amount of load can be locally satisfied by

TABLE VI
OPTIMAL PLANNING STRATEGIES FOR DERS IN THREE SCENARIOS

Scenario	Node 4			Node 5		
	WT	PV	ESS	WT	PV	ESS
S1	0	0	0	0	0	0
S2	3	2	2	0	0	2
S3	3	2	2	2	0	2

TABLE VII
OPTIMAL PLANNING STRATEGIES FOR TNS IN THREE SCENARIOS

Line No.	8	9	10	11	12	13	14	Investment (million $)
S1	0	1	0	1	1	1	1	53
S2	1	1	1	1	0	0	0	42
S3	1	1	1	0	0	0	0	32

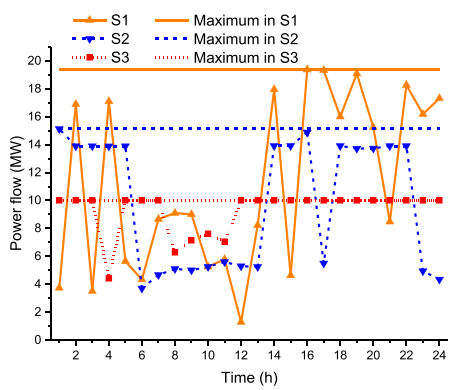

Fig. 7. Maximum power flows on line 3–5 in three scenarios.

DERs. In S3, only 3 transmission lines, i.e., No. 8, 9 and 10, are optimized for construction in S3. However, 5 and 4 lines are respectively optimized for construction in S1 and S2. Therefore, transmission prices can further encourage users to invest in DERs, which potentially defers transmission expansion and reduces the system-wide investment costs.

In Table VII, line 3–5, i.e., Line No. 11, is removed from the planning strategy in S3. The maximum power flows on line 3–5 in three scenarios are compared in Fig. 7.

As one can observe, the maximum power flows can be effectively limited below the transmission capacity of 10 MW in S3. However, with no or less DERs, more transmission lines have to be constructed to satisfy the load in S1 and S2. The maximum power flows reach 19.38 MW and 15.13 MW in S1 and S2, respectively. The costs of the users at nodes 4 and 5 in three scenarios are compared in Table VIII.

From the results in Table VIII, the transmission costs in S3 account for 8.84% and 19.77% of the total costs at nodes 4 and 5, respectively. Thus, transmission cost is an important component of users' electricity cost. The iteration between the users at nodes 4 and 5 is explained as follows: Firstly, the users at node 5 find it beneficial to invest in DERs. After investing in DERs at node 5, the users at node 4 will find an increase in his/her total costs

TABLE VIII
COSTS OF THE USERS AT NODE 4 AND 5 IN THREE SCENARIOS (UNIT: MILLION $)

	S1		S2		S3	
	4	5	4	5	4	5
Investment cost	0	0	1.88	0.02	1.88	1.02
Energy cost	14.09	9.07	17.85	6.39	16.89	5.15
Transmission cost	2.07	1.81	1.86	1.81	1.82	1.52
Total cost	16.16	10.88	21.59	8.22	20.59	7.69

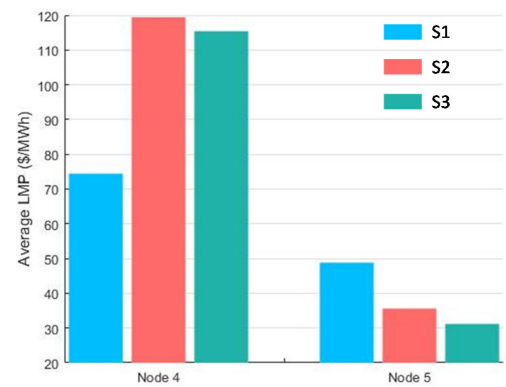

Fig. 8. Daily average LMPs at node 4 and 5 in three scenarios.

because of the change in LMPs and transmission prices caused by the investments at node 5. Then, the users at node 4 are driven to invest in DERs to reduce the electricity costs. The iteration continues until equilibrium is achieved.

By comparing the results in S3 with those in S1, the total cost of the user at node 5 can be greatly reduced, while the total cost of the user at node 4 will increase after DERs investment. This is because 3 transmission lines from node 4 to node 6 are optimized for construction in S1, which are absent in S3. These transmission lines will significantly alleviate the power congestion around node 4, thereby reducing the LMPs at node 4. The daily average LMPs at node 4 and 5 in three scenarios are compared in Fig. 8. As one can observe, the daily average LMP at node 4 reaches 115.44 $/MWh in S3, increasing by 55.12% of that in S1. However, the daily average LMP at node 5 decreases with the investments in DERs. Thus, the total costs of the user at node 4 increase, while the total costs of the user at node 5 decrease in S3.

Therefore, the impacts of TCA on the co-planning for TNs and DERs are threefold: i) TCA can reflect different users' actual usage of transmission assets, thereby leading users to investing in DERs. ii) Power flows of transmission lines can be effectively reduced; iii) The DERs can substitute transmission lines for locally satisfying load, which helps defer transmission expansion and reduce the system-wide investment costs.

3) Impacts of DERs' Investment Costs: In this subsection, we conduct the sensitivity analysis on the annualized investment costs for WTs, PVs and ESSs. With the decrease in the costs for DERs in the future, our results offer an analysis for the impacts of TCA on co-planning for TNs and DERs. Given the annualized

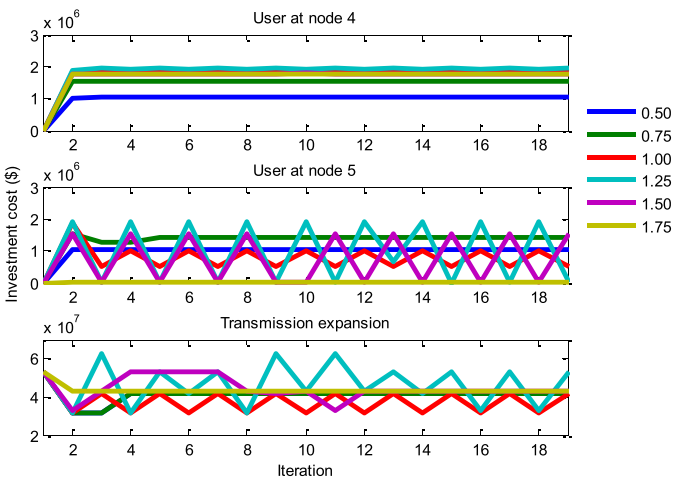

Fig. 9. The iterative total investment costs under different cost rates.

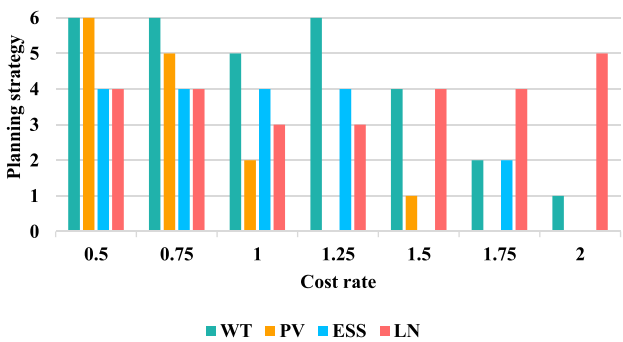

Fig. 10. Desired solutions to the optimal strategies under different cost rates.

investment costs as the base case, a cost rate is introduced to increase or decrease the investment costs for a single DER. Under different cost rates, the iterative total investment costs are shown in Fig. 9.

From the results, when the investment costs for a single DER are set to 0.50, 0.75 and 1.75 times of those in the base case, a stationary equilibrium can be obtained. However, when the cost rates are set to 1.00, 1.25 and 1.50, the planning decisions for DERs and TNs oscillate.

In the cases with the cost rates 1.00 and 1.25, one can observe that the dilemma illustrated in Section IV-C occurs: the user at node 5 invests in the marginal DERs, replacing the marginal transmission line. However, the reduction in electricity and transmission costs cannot recover the investments for the marginal DERs. Therefore, the user will not invest in the marginal DERs, leading to the expansion for the marginal transmission line. In the case with the cost rate 1.50, only the planning strategy of the user at node 5 oscillates, while the strategies of the user at node 4 and transmission expansion are stationary.

Our proposed criterion for desired solutions works for the oscillated cases. By comparing the system-wide total costs for the oscillated strategies, the desired solutions to the optimal strategies under different cost rates are shown in Fig. 10.

With the cost rate increasing from 0.50 to 2.00, the investments for WTs, PVs and ESSs gradually decrease from 6 to 1, from 6 to 0 and from 4 to 0, respectively. To satisfy the transmission capacity requirements, the investment for transmission lines increase from 3 to 5.

TABLE IX
OPTIMAL PLANNING STRATEGIES FOR TNS AND DERS UNDER DIFFERENT DERS' VARIABLE COSTS

Variable cost	0	1 $/MWh	5 $/MWh	10 $/MWh
Node 4	3 WTs 2 PVs 2 ESSs	3 WTs 2 PVs 2 ESSs	3 WTs 2 PVs 0 ESSs	0 WTs 0 PVs 0 ESSs
Node 5	2 WTs 0 PVs 2 ESSs	0 WTs 0 PVs 0 ESSs	0 WTs 0 PVs 0 ESSs	0 WTs 0 PVs 0 ESSs
TNs	3	4	4	5

4) Impacts of DERs' Variable Costs: In practice, the variable costs of DERs including the operation and maintenance costs may influence the planning strategies for TNs and DERs. Therefore, in this subsection, we analyze the impacts of DERs' variable costs.

To incorporate DERs' variable costs, the objective function (11) is modified as follows:

$$\min \; C_x^{DER} + C_x^{LD} + C_x^{V}, \tag{41}$$

where C_x^V is defined as the variable costs of DERs:

$$C_x^V = \sum_{s=1}^{N^S} \gamma_s \sum_{t=1}^{T} \left[\sum_{i=1}^{N^{WT}} c_i^{WT} P_{i,t,s}^{WT} + \sum_{i=1}^{N^{WT}} c_i^{PV} P_{i,t,s}^{PV} + \sum_{i=1}^{N^{WT}} c_i^{ESS} \left(P_{i,t,s,\alpha}^{ESS} + P_{i,t,s,\beta}^{ESS} \right) \right] \Delta T, \tag{42}$$

where c_i^{WT}, c_i^{PV} and c_i^{ESS} are the variable costs of WTs, PVs and ESSs, respectively. As shown in [38], the variable costs of DERs are generally from 0.1 cents/kWh to several cents/kWh. Table IX shows the planning strategies for TNs and DERs under different DERs' variable costs.

As one can observe, with the increase in DERs' variable costs, the investments for DERs at nodes 4 and 5 decrease because the profits from DERs cannot cover the costs for DERs. As a result, more TNs are required to satisfy the load demands.

B. IEEE 118-Bus System

In the IEEE 118-bus system, five nodes and ten candidate transmission lines are selected for illustration. The users at nodes 2, 13, 14, 16 and 117 make investments in DERs. Each user has an annualized budget of $2,600,000 for candidate DERs, including 5 WTs, 4 PVs and 3 ESSs. Note that the sites for DERs investment are pre-defined in this paper. The optimal siting and sizing for DERs considering TCA deserves an in-depth study in the future. The parameters of the candidate transmission lines are shown in Table X.

To maximize the annualized revenues, the users at different nodes will strategically plan for and use the DERs. The planning strategies for DERs at five nodes in S2 and S3 are compared in Table XI.

TABLE X
PARAMETERS OF CANDIDATE TRANSMISSION LINES IN IEEE 118-BUS SYSTEM

Line No.	From	To	Reactance (p.u.)	$P^{LN,\max}$ (MW)	Investment (million $)
1	2	12	0.062	10	10
2	1	2	0.100	10	10
3	13	15	0.244	10	10
4	11	13	0.073	10	10
5	14	15	0.195	10	10
6	12	14	0.071	10	10
7	16	17	0.180	10	10
8	12	16	0.083	10	10
9	12	117	0.140	10	10
10	12	117	0.140	10	10

TABLE XI
OPTIMAL PLANNING STRATEGIES FOR DERS IN S2 AND S3

Node No.	S2			S3		
	WT	PV	ESS	WT	PV	ESS
2	4	2	0	4	2	3
13	4	3	3	4	3	3
14	3	3	3	4	3	3
16	4	3	3	4	3	3
117	3	4	0	3	4	3

TABLE XII
OPTIMAL PLANNING STRATEGIES FOR TNS IN THREE SCENARIOS

No.	1	2	3	4	5	6	7	8	9	10
S1	1	0	1	1	0	0	1	0	1	1
S2	0	0	1	1	0	0	1	0	1	0
S3	0	0	1	1	0	0	1	0	0	0

In S3, 14 iterations and 593.60 s are taken to reach equilibrium. Note that no DERs are planned in S1. By considering TCA in the planning model, users will be economically driven to invest in more DERs. Compared with the results in S2, 3 more ESSs are planned at node 2 and 117, and 1 more WT is planned at node 14 in S3. Table XII shows the planning strategies for TNs in three scenarios.

As one can observe, there are 6 transmission lines in S1, 4 transmission lines in S2 and 3 transmission line in S3. The results demonstrate that the additional DERs can further defer the planning for TNs. In Table XII, lines 12–117, i.e., Line No. 9 and 10, are removed from the planning strategy in S3. The maximum power flows on line 12–117 in the three scenarios are compared in Fig. 11.

Without DER in S1, the maximum power flow on line 12–117 reaches 24.48 MW, which requires 2 lines from nodes 12 to 117. In S2, 3 WTs and 4 PVs are invested in, and the maximum power flow on line 12–117 is reduced to 10.45 MW, which only requires 1 candidate line, i.e., line 12–117. By considering TCA in the planning of DERs, in S3, 3 more ESSs are invested in compared with the results in S2. Then, the maximum power flow on line 12–117 can be further reduced to 8.68 MW. Thus,

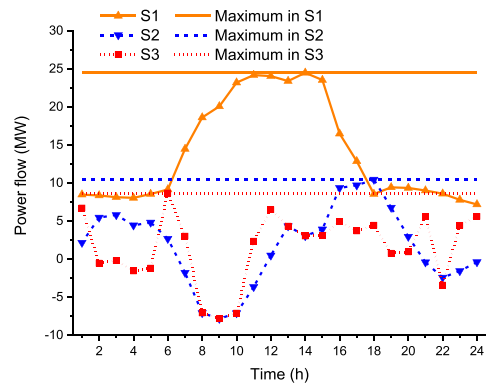

Fig. 11. Maximum power flows on line 12–117 in three scenarios.

the maximum power flow can be effectively reduced, thereby deferring the transmission expansion.

VI. CONCLUSION

In this paper, a tri-level expansion planning model for TNs and DERs considering TCA is proposed. The interactions between energy users and market operator are formulated as an EPEC, which is approached by a diagonalization method. An iterative algorithm is proposed to solve the tri-level model, and the convergence analysis is conducted. Case studies based on a modified Garver's 6-bus system and the IEEE 118-bus system demonstrate that i) a well-designed TCA scheme can reveal different users' actual usage of transmission assets, ii) hourly transmission prices can incentivize users to strategically invest in DERs, and iii) hourly transmission prices can help reduce power flows on transmission lines, thereby deferring transmission expansion. Therefore, it is imperative to incorporate TCA in the expansion planning for TNs and DERs.

In our future work, two issues deserve an in-depth study: i) A novel TCA scheme should be designed to reflect the DERs' contributions to reducing peak power flows on transmission lines. In practice, DERs can support the secure operation of power grids during peak load hours by reducing power flows on transmission lines. An incentive-compatible pricing mechanism should be designed for this grid-friendly manner. ii) An optimal placement model for DERs considering distribution networks and voltage regulation should be developed.

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Jianxiao Wang (S'14) received the B.S. degree in electrical engineering in 2014 from Tsinghua University, Beijing, China, where he is working toward the Ph.D. degree in electrical engineering. He is also a Visiting Student Researcher at Stanford University, Stanford, CA, USA. His research interests include power system operation and planning, and electricity market.

Haiwang Zhong (S'10–M'13) received the Ph.D. degree in engineering from Tsinghua University, Beijing, China, in 2013. He is currently an Associate Professor with Tsinghua University. His research interests include generation scheduling optimization, demand response, and electricity markets.

Wenyuan Tang (S'14–M'17) received the B.Eng. degree in electrical engineering from Tsinghua University, Beijing, China, in 2008, and the M.S. degree in electrical engineering, the M.A. degree in applied mathematics, and the Ph.D. degree in electrical engineering from the University of Southern California, Los Angeles, CA, USA, in 2010, 2014, and 2015, respectively.

He is an Assistant Professor with the Department of Electrical and Computer Engineering, North Carolina State University, Raleigh, NC, USA. Prior to joining the NC State faculty in 2017, he was a Postdoctoral Scholar with the University of California, Berkeley, and Stanford University. His research interests include power system operation, energy economics, and data analytics for smart grids.

Ram Rajagopal received the M.A. degree in statistics and the Ph.D. degree in electrical engineering and computer sciences from the University of California Berkeley, Berkeley, CA, USA. He is an Associate Professor of civil and environmental engineering and electrical engineering (by Courtesy) with Stanford University, Stanford, CA, USA, where he directs the Stanford Sustainable Systems Lab, focused on large-scale monitoring, data analytics, and stochastic control for infrastructure networks, in particular power and transportation. He holds more than 30 patents from his work, and has advised or founded various companies in the fields of sensor networks, power systems, and data analytics. His current research interests in power systems are in integration of renewables, smart distribution systems, and demand-side data analytics. He has also built wireless sensing systems and large-scale analytics for transportation and power networks. He was a recipient of the Powell Foundation Fellowship, Berkeley Regents Fellowship, the NSF CAREER Award, the Makhoul Conjecture Challenge award, and best paper awards in various conferences.

Qing Xia (M'01–SM'08) received the Ph.D. degree in engineering from Tsinghua University, Beijing, China, in 1989. He is currently a Professor with Tsinghua University. His research interests include electricity markets, generation scheduling optimization, and power system planning.

Chongqing Kang (M'01–SM'07–F'17) received the Ph.D. degree in engineering from Tsinghua University, Beijing, China, in 1997. He is currently a Professor with Tsinghua University. His research interests include power system planning, power system operation, renewable energy, low carbon electricity technology, and load forecasting.